

DEEPAKRAJ B

FULL STACK ENGINEER | REACT, TYPESCRIPT, PYTHON, JAVA & CLOUD

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PROFILE

Full Stack Software Engineer with hands-on experience building production web applications, interactive user interfaces, and scalable backend microservices. Skilled across the modern frontend stack with **React, TypeScript, JavaScript, HTML5/CSS3**, component-driven design, and client-side state management, paired with robust backend APIs using **Python (FastAPI, Django)** and **Java (Spring Boot)**. Proven track record owning systems end-to-end - from responsive UI dashboards and field-ready mobile interfaces to relational database architecture (**PostgreSQL, MySQL, Redis**) and containerized cloud deployment on **AWS (EC2, S3, Docker)**. Experienced delivering on-site under strict operational constraints with offline-first reliability.

PROFESSIONAL EXPERIENCE

Software Engineer | Kritilabs Technologies Pvt. Ltd., Chennai, India July 2024 - Present

Sentinel Kote - Offline RFID Weapon Tracking System (Indian Army)

React, Android, Java, REST APIs, Offline State, RFID

- Architected and delivered the full-stack offline platform solo within 2 weeks, deploying on-site at an **Indian Army camp** to replace error-prone manual paper ledger workflows.
- Engineered the **React inventory web dashboard** with modular reusable components, stateful form validation, and optimistic UI updates for real-time weapon check-in/out, status filtering, and audit log generation.
- Designed an intuitive touch UI for **Android handheld RFID terminals** with instant visual scan feedback and implemented **offline-first local caching** with Java REST APIs, ensuring zero data loss without internet.

AI Natural Language Query Engine & Interactive Dashboard

React, TypeScript, Python, FastAPI, LangGraph, Docker, AWS

- Built an end-to-end full-stack query platform connecting a responsive **React/TypeScript single-page application** to a high-throughput **Python / FastAPI** backend for natural-language database querying.
- Engineered interactive UI components featuring real-time query progress streams, SQL syntax highlighting, schema metadata exploration, and dynamic tabular result visualization.
- Orchestrated multi-step prompt validation and vector schema retrieval using **LangGraph and ChromaDB**; containerized services with **Docker** and established automated **AWS CI/CD pipelines**.

Enterprise ERP Web Platform & Client Integration Portals

React, JavaScript, Python, Frappe, REST APIs, Webhooks, MySQL

- Engineered full-stack features and client portal views for an ERPNext platform serving 700+ enterprise users across 4 manufacturing units, managing BOMs, inventory workflows, and work orders.
- Developed **REST API integrations and webhook receivers** connecting React-based client interfaces, inventory databases, and mobile barcode scanners with Frappe APIs, reducing manual data entry by **80%**.
- Optimized **MySQL & PostgreSQL** database schemas, query indexes, and API payload serialization, improving REST API response times by **30%**.

Real-Time Fleet & Security Telemetry Monitoring Platform

React, Java, Spring Boot, WebSockets, Redis, Leaflet, JWT

- Developed real-time monitoring web dashboards with interactive map visualizations and live **WebSocket** data streams for live vehicle fleet tracking and smart lock telemetry for **IOCL and BPCL**.
- Implemented secure authentication with **JWT and OAuth2** dynamic RBAC across frontend routes and Java backend microservices, and deployed analytics services for **Indian Railways**.

TECHNICAL SKILLS

Frontend Development: React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Component Architecture, State Management

Databases & Caching: PostgreSQL, MySQL, Redis, SQLite, SQLAlchemy, Database Schema Design, Query Optimization

Systems & Security: Full-Stack Architecture, Offline-First Systems, JWT, OAuth2, RBAC, Event-Driven Architecture, Unit Testing

Backend Engineering: Python (FastAPI, Django, Flask), Java (Spring Boot), RESTful APIs, WebSockets, Webhooks, Microservices

Cloud, DevOps & Tools: AWS (EC2, S3), Docker, CI/CD Pipelines, Linux, Nginx, Git, Postman, Vite, Swagger/OpenAPI

AI & Query Integration: LangGraph, LangChain, NL2SQL, OpenAI APIs, ChromaDB, Prompt Engineering, Structured Outputs

EDUCATION

B.Tech in Computer Science & Engineering | SASTRA Deemed University, Thanjavur, Tamil Nadu **CGPA: 7.2955** | 2020 - 2024

CERTIFICATIONS

- Backend Development Specialization - Meta
- Software Engineering Specialization - IBM
- Python Crash Course - Google
- Machine Learning Specialization